

But Microsoft CEO Satya Nadella put a happy spin on the whole episode, citing a 160-year-old economics concept to suggest that this was good news.

"Jevons paradox strikes again!" Nadella wrote on social media, sharing the concept's Wikipedia page. "As AI gets more efficient and accessible, we will see its use skyrocket, turning it into a commodity we just can't get enough of."

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In other words, don't worry, investors! This obscure economics concept shows a profitable path forward even though our competitors are drastically undercutting us. Look, here's Wikipedia and a story about 19th-century coal.

Jevons paradox may remain somewhat obscure, but over the last few decades, it has been cited as a reason why more energy-efficient cars, appliances and light bulbs may fail to reduce our consumption of fossil fuels. It has been cited as a reason why building more highway lanes fails to solve traffic congestion. And it's being cited by people in the AI realm — beyond just Nadella — as a potential reason why AI may actually create more jobs in some occupations rather than mass layoffs.

Today in the *Planet Money* newsletter, what is Jevons paradox? Where does it come from? How much of it is BS? And why does this 160-year-old concept all of a sudden seem relevant for our futuristic age of AI?

The origin of Jevons paradox

In the early 1860s, English leaders began to worry about coal. The combustible black rock found deep in the ground had literally powered the Industrial Revolution. England's economy was completely dependent on it. And now leaders were worried it was going to run out.

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Enter economist William Stanley Jevons. Born in Liverpool — like The Beatles! — Jevons was likewise on his way to international stardom. Before Beatlemania, there was Jevonsmania.

OK, fine, so Jevons was never really that popular. Only nerds have ever known or cared about him.

Nonetheless, back in the mid-19th century, Jevons was pioneering the use of math and statistics in economics, charting the way for what would become known as neoclassical economics, which would dominate the field for centuries to come.

When English leaders began fretting about coal, the late-20-something Jevons decided he wanted to address their concerns. And so, in 1865, he published a book named *The Coal Question: An Inquiry Concerning the Progress of the Nation, and the Probable Exhaustion of Our Coal-Mines*.

At the time, the English might have taken comfort in the idea that their coal-shortage problem could be solved, or at least punted down the road, by technological innovations that increased the efficiency of coal consumption. Surely, all it would take was building engines and furnaces that produced more power with less coal. More-efficient technologies would reduce coal consumption, and then maybe the coal-less doomsday could be averted for a long time.

Jevons argued that, basically, this solution was dumb. He spent an entire chapter in *The Coal Question* arguing that more-energy-efficient technology could not solve England's coal-shortage problem. In fact, Jevons wrote, greater efficiency of coal-powered technologies would — paradoxically — lead to *more* consumption of coal, not less.

"It is wholly a confusion of ideas to suppose that the economical use of fuel is equivalent to a diminished consumption," Jevons wrote. "The very contrary is the truth."



William Stanley Jevons
via Getty Images

Jevons' thinking was this: With greater efficiency, you get more for less, more power out of every lump of coal. And that effectively lowers the cost of coal energy. You can now burn your furnace or run your factory's machines longer for cheaper. And, lower prices mean higher demand.

And we're not talking just like a bit more demand. Jevons was living during a time in which coal was fueling rapid industrialization. When he looked around him, he saw factories being built, infrastructure getting erected and new coal-powered inventions radically changing his society. When he saw engines and furnaces getting more efficient and thus cheaper to run, he saw that as sparking an atomic explosion of demand.

As coal-powered steam engines became cheaper to run, for example, companies and governments found all sorts of new applications for the technology. Jevons uses all sorts of old-timey examples. The "cheapness of the power," he wrote, "allows us to draw rivers from our mines" and "drive our coal-pits." More-efficient steam engines were deployed in new factories, trains and steamships.

Likewise, more-efficient blast furnaces, which used heat from coal to smelt iron, helped the iron industry become more profitable and grow bigger. Building even more-efficient furnaces, he wrote, would only help grow the industry more — and with it the number of furnaces it used. That would "more than make up for the diminished consumption of each."

In the macro view, Jevons suggested, more-efficient use of coal would stimulate economic growth, increasing coal consumption across the economy. "It is the very economy of the use of coal that makes our industry what it is," Jevons wrote, "and the more we render it efficient and economical, the more will our industry thrive, and our works of civilization grow."

If anything, Jevons claimed, making coal-powered technologies more energy efficient would only speed up the inevitable collapse of coal resources. Greater efficiency increased consumption, not decreased it. This was the original iteration of what became known as the Jevons paradox.

The resurrection of Jevons paradox

Despite *The Coal Question* becoming a bestseller in the 19th century and despite Jevons' historic contributions to economics, the Jevons paradox was mostly forgotten about in the 20th century. That is until the 1970s, when the modern environmental movement and an oil crisis led economists to start thinking hard

about energy efficiency again. Governments began pursuing policies like mandating greater fuel efficiency for cars, and people began to wonder, "How much will that really reduce our consumption of fossil fuels?"

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It was within this context that economists rediscovered the Jevons paradox. And they created a modern formulation that's a bit more nuanced. The idea is that making things like cars and appliances more energy efficient creates a "rebound effect." When you make a machine more energy efficient, it effectively lowers the cost of using it. And — hello, the classic law of demand from economics — when stuff gets cheaper, people tend to use or consume more of it.

So, for example, with more-fuel-efficient cars, it gets cheaper to travel every mile, so people drive more miles. Some may decide to stop riding the bus and buy a car. Some families may buy a second car. Others may buy bigger vehicles, like SUVs. With more-efficient light bulbs, people may keep their lights on for longer or build things like the Sphere in Las Vegas.



The Sphere in Las Vegas.

Anadolu/Getty Images

The rebound effect from greater energy efficiency reduces the environmental benefits of energy-efficient technologies.

Now, there's nothing paradoxical about stuff getting cheaper and demand going up. That's classic economics. And if the rebound effect is small, then gains in energy efficiency will still lead to a reduction in overall energy consumption. There is no real Jevons paradox unless energy demand dramatically explodes in response to the lower energy costs of more-efficient machines.

There has been a lot of debate over how big rebound effects actually are in various energy markets. As far as we can tell, it seems like most economists who analyze this closely find that, generally speaking, rebound effects in modern energy markets tend to be rather small. They're like, sure, the rebound effect is real and it dilutes the environmental benefits of greater energy efficiency, but it doesn't come close to completely canceling them out. So, for example, if I buy a Prius, maybe I'll go on more road trips or whatever, but that doesn't negate all the fuel savings over the vehicle's lifetime. Studies suggest that energy-efficient vehicles and appliances still, on net, result in lower energy consumption.

But others stress that the ripple effects of more-energy-efficient technologies are harder to measure. They argue that, when you look at the bigger picture of how demand explodes in response to cheaper energy, estimates of rebound effects have been too small. In some cases, they argue, the rebound effect can be so big that increasing something's efficiency can backfire, resulting in greater overall energy consumption — the very paradox that Jevons wrote about.

In a 2010 article in *The New Yorker*, "The Efficiency Dilemma," journalist David Owen made a big deal out of the Jevons paradox in the modern world. He pointed to what happened after refrigerators became more energy efficient during his lifetime. Some Americans began buying multiple refrigerators for their homes. Hotels began putting mini-fridges in every room. "Gas stations now often have almost as much refrigerated shelf space as the grocery stores of my early childhood," he wrote. More-efficient refrigeration also encouraged people to buy more food, because they could keep it longer, and that often leads to food waste, another energy cost. This refrigeration technology was also adopted in air conditioners, and those have similarly proliferated in cars, homes and buildings around the world. All of this, Owen suggests, has resulted in a massive rebound effect where maybe more-energy-efficient refrigeration technology has actually resulted in overall higher energy consumption.

However, U.K. journalist Tim Harford (a frequent guest of *Planet Money*) writes that Owen blows this all out of proportion. Sure, there are real rebound effects. But, he writes, "In my lifetime, energy consumption per person in the UK has fallen by one-third, while carbon dioxide emissions per person have fallen by nearly 60 per cent."

In short, the extreme version of Jevons paradox in today's energy markets may be BS. Very likely, greater efficiency does not lead to greater consumption of fossil fuels. But maybe that's because today's economy is unlike the economy of Jevon's lifetime, when England was experiencing rapid, coal-powered industrialization. These days, gains in efficiency and resulting reductions in energy costs don't result in the same explosion of energy demand.

Jevons paradox and AI

All of a sudden, it seems like a lot of people are talking about the potential for a Jevons paradox in the market for artificial intelligence.

This wouldn't be the first time the concept had jumped from its traditional home in energy economics. Over the years, scholars and writers have seen versions of it in other places. For example, scholars have found evidence of a Jevons paradox in water markets, where, for example, greater efficiency in irrigation leads to greater overall water consumption. Others have found evidence of a Jevons paradox for infrastructure, where, for example, more highway lanes end up creating more traffic congestion.

And now people like Microsoft's Nadella are embracing the concept, suggesting he believes — or he wants *us* to believe — that as AI gets more efficient and cheaper, demand for it will explode and, despite bruising competition from companies like DeepSeek, there will be many profits to be made, including by Microsoft and its business partner OpenAI. That is far from guaranteed. We still don't know what AI will be capable of, how consumers will ultimately want to use it, and what competition, costs and a host of other factors will look like in the AI market.

But the Jevons paradox may also be relevant in the AI realm for another reason: It could help us understand what could happen to some occupations if AI revolutionized them and made workers more productive.

Last month, *Planet Money* attended the annual meeting of American economists, and we saw Stanford University economist Erik Brynjolfsson talk about the Jevons paradox and what it could mean for certain occupations.

As previously explained, Jevons saw that as coal-powered tech got more efficient, it led to an increase in demand for coal, not a decrease. Brynjolfsson sees the potential for something similar with certain occupations and AI. That is, as AI makes certain occupations more efficient, it could lead to an increase in demand for human labor, not mass layoffs.

Brynjolfsson points to what happened to airplane pilots after jets were invented as an example of what could possibly happen in some occupations. "Pilots became dramatically more productive and effective once jets were invented," Brynjolfsson says. They could travel farther and faster, effectively lowering the cost of every mile of air travel. "Did that mean that we didn't need as many pilots because now pilots could do more work? No. We consumers decided that we're going to fly more than ever. So now a lot more people fly. And there's more demand for pilots."

Now, for more clarity, a Jevons paradox for AI-affected occupations will only really happen if at least three crucial boxes are checked:

First, workers get more productive thanks to AI. That's not guaranteed. On the one hand, AI could turn out to be really amazing and not need humans to do certain occupations at all. On the other hand, AI could prove to be largely a dud for productivity — kinda like the smartphone has so far proved to be — and not really boost how much workers can produce per hour.

Second, higher worker productivity translates into lower prices. This doesn't mean that workers have to get paid less. It means that as they make more widgets or fly more miles or generate more ideas per hour, the prices of what they're selling to consumers effectively drops.

Third, and this is really crucial, consumer demand needs to explode in reaction to lower prices. (To use econ jargon, this hinges on whether demand is "elastic" or "inelastic.")

Brynjolfsson believes there will be certain occupations that check each box. In fact, he suggests, there may already be some occupations seeing a Jevons paradox-like process unfold. For example, coders, translators and radiologists are all occupations that may be seeing AI increasing their productivity and that seem to be growing despite (or because of) that.

That's the happy scenario, where a Jevons paradox makes workers more efficient and increases demand for their labor, rather than decreasing it. There's lots of job growth, and wages may even go up.

Of course, technological change won't do this for all occupations. For one, if your employer doesn't share the fruits of higher productivity with you or if the machine makes your job easier to do and more people can do it, you might not see higher wages. You could see falling wages over time. And, of course, if a machine can completely automate every task of your job, you're out of luck. And even if it doesn't, and if humans are still needed to do the job, massive gains in productivity can still shrink the number of workers needed in some occupations.

Take, for example, farmer jobs. Around 1900, 40% of American workers worked in agriculture. Today, it's less than 2%.

Why did the percentage of Americans farming for a living radically shrink? For one, new technologies, like tractors and robotic fruit and vegetable pickers, dramatically increased farmer productivity in making food. But — and this is crucial — Brynjolfsson says demand for food didn't explode as much as was needed to keep mass numbers of farmers employed (in other words, demand for food proved to be "inelastic").

"A few farmers can produce way more food than they did before, and however much as we try, we just aren't eating that much more," Brynjolfsson says. In other words, the price of food has fallen as farmers have gotten more efficient, but the rebound effect — the increase in demand in response to falling food prices — has not exploded enough to offset the job-killing effects of each farmer being able to produce a lot more food. There's a relatively fixed amount of food we eat, and with farmers able to produce much more in the modern era, we don't need as many farmers to satisfy food demand.

But Brynjolfsson is a self-declared "techno-optimist." While he recognizes that automation has created losers throughout history, including in big sectors like farming and manufacturing, he sees technological change as being a net win for jobs in the long sweep of history.

"We've had a lot of amazing technologies — bulldozers and computers and spreadsheets and all sorts of other things — that made people radically more efficient in what they're doing," Brynjolfsson says. And yet, perhaps paradoxically, "We just keep hiring more and more people."

That said, we don't really know what AI is going to be capable of, and, Brynjolfsson says, he doesn't rule out the possibility that the future could be different.

Interestingly, Jevons himself — in that very same chapter in *The Coal Question* where he outlines his paradox — talks specifically about his paradox in action in the labor market.

After "the introduction of new machinery" increases worker productivity, Jevons wrote, it often "throws labourers out of employment for the moment. But such is the increased demand for the cheapened products, that eventually the sphere of employment is greatly widened. Often the very labourers whose labour is saved find their more efficient labour more demanded than before."

Of course, predicting the future of technological change is very hard. In fact, William Stanley Jevons is a cautionary tale for this. In retrospect, Jevons was embarrassingly wrong in *The Coal Question*. He predicted that England would indeed run out of coal, which would create a serious problem for its economy. He advised that, basically, England should stop deficit spending and brace for the inevitable collapse of its economy. What he failed to predict: The nation would find other sources of energy to power its industry in the years to come.